

The Role of Employers in Phased Retirement

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Abstract

Phased retirement allows older workers to combine part-time wages and pensions and to remain productive until or beyond the legal retirement age. It allows organizations to retain valuable workers that can gradually transfer their skills to younger colleagues. Although worker preferences are widely studied, the discrepancy between stated preferences and actual usage suggests employer restrictions play a significant role. This paper adds to the scarce literature on employers by analyzing survey data from human resource managers responsible for organizational personnel policy. We find that while few organizations completely deny phased retirement, in line with the Dutch institutional context, formal arrangements covering all employees are rare. Instead, most employers offer phased retirement selectively, depending on the nature of the job and worker characteristics. For example, the productivity of older workers relative to younger colleagues matters, and phased retirement opportunities are less common in jobs where part-time work is less prevalent.

JEL codes: J14, J23, J26

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1. Introduction

Phased retirement refers to older workers gradually reducing hours and effort while staying in the same organization, possibly combining earnings with a partial pension to keep income close to full-time levels. The advantages of phased retirement versus an immediate transition to full retirement (“cliff edge” retirement) are well-documented (e.g., Rogerson and Wallenius, 2020). For the economy, phased retirement can increase labor force participation and potentially total hours worked, if the participation effect dominates the intensive-margin effect (Ameriks et al., 2020). It can also contribute to pension sustainability if individuals pay premiums longer and claim pensions later. Analyzing demand, supply, and institutional drivers of phased retirement can therefore inform policies to increase economic activity and engagement of older age groups.

Prior research has mainly focused on workers’ choices regarding phased retirement, with much less on whether and to whom organizations offer it. This study contributes to understanding employer strategies concerning phased retirement, as part of broader flexible work arrangements (FWAs; Hill et al., 2008) aimed at retaining older workers, increasing engagement, and raising productivity.

For workers, phased retirement has several advantages. As Hutchens and Grace-Martin (2006) note, it can “produce a more fulfilling end to a lifetime of work.” Compared with full-time work, it is less stressful and demanding, while earnings fall less than with full retirement. It allows workers to gradually develop new activities and adjust to a life less dominated by paid work, and to build a social network that replaces work contacts. It also entails lower transition costs than other gradual-retirement forms, such as bridge jobs with a new employer or self-employment, typically involving part-time and less challenging work.

Survey data therefore often show a preference for phased over abrupt retirement, but stated interest exceeds actual use. Obvious explanations include employer restrictions and institutional constraints. In some pension systems, defined-benefit pensions depend on final earnings without a part-time correction, making part-time work shortly before retirement less attractive. Pension regulations may also hamper combining earnings and pension benefits from the same employer.

This paper focuses on how employers handle phased retirement, which has been studied much less than employee preferences. For employers, phased retirement has both advantages and drawbacks. The main advantage is retaining older workers with valuable expertise who can gradually transfer knowledge to younger colleagues, potentially “increasing productivity through preservation of firm specific human capital” (Hutchens and Grace-Martin, 2006). However, phased retirees and part-time employees also entail costs, such as fixed per-employee costs independent of hours, leading some employers to impose minimum-hours constraints. Phased retirement and part-time jobs may be unattractive where part-time work is hard to organize, for example due to specific occupational schedules. Instead of generic policies, employers may offer phased retirement selectively, which can increase inequality and disadvantage vulnerable workers with low earnings or poor health. Kossek and Kelliher (2023) therefore argue that individualized flexible-work deals should be shaped at the firm or sector level.

The goal of this paper is to get more insight into why some employers and organizations in the Netherlands facilitate phased retirement more than others, to whom, and why. We analyze new Dutch survey data on employers and human resource managers, who were directly asked

about their establishment's phased retirement policies and practices. The survey is modelled after the employer survey designed by Hutchens and Grace-Martin (2006) and Hutchens (2010) for the US, and the main purpose is to investigate how employers handle phased retirement within a country with a very different institutional setting. In particular, the Netherlands provides an ideal setting for this investigation because its institutional framework subsidizes and protects the supply side of gradual retirement. Dutch workers possess a statutory right to request a reduction in working hours that employers must honor unless they can prove compelling business interests, while occupational pension funds allow flexible, partial drawdown of accrued pension rights to supplement part-time wages and smooth consumption. Moreover, unlike systems with a Social Security Earnings Test that penalize continued work and induce abrupt retirement, the Dutch system features no earnings test to discourage part-time employment. Because these formal frameworks ensure that supply-side participation is not bottlenecked by institutional constraints, the Dutch context substantially helps to isolate the role of the firm, leaving room for remaining limitations on actual usage to apply primarily on the demand side. The results we find are surprisingly well in line with the earlier findings for the US.

We start with a brief literature review that suggests that phased retirement has clear advantages for workers' vitality. Nonetheless, there is mixed evidence on the net labor supply effects of phased retirement, largely depending on the institutional context. Even for the Netherlands, some studies find negative and other studies find positive effects of phased retirement on total labor supply of older age groups. The large discrepancy between the stated willingness of workers to reduce working hours in the years before retirement and the actual use of phased retirement suggests demand-side obstacles.

Our survey data suggest that few Dutch organizations do not offer phased retirement at all, as might be expected given the Dutch institutional context that aims at facilitating workers who desire to reduce their hours of work. On the other hand, not every worker has access to phased retirement - many employers offer phased retirement options to a selective group of employees only, depending on the nature of the job and on worker characteristics. These findings largely corroborate the existing findings. For example, productivity of older workers compared to their younger colleagues matters, and opportunities for phased retirement are less common for jobs where part-time work in general is not prevalent. A formal phased retirement policy creates more phased retirement opportunities than informal or ad hoc arrangements. However, such formal policies are less common in firms characterized by a larger share of 55+ workers, and if they are, they often allow for exceptions.

The remainder of this paper is organized as follows. Section 2 gives an overview of the existing literature on phased retirement, focusing on the role of employers. Section 3 describes the relevant institutional context in the Netherlands and uses a second source of survey data, on individual workers, to analyze the prevalence of (intended) phased retirement among current workers and retirees. In Section 4, we analyze data on organizational attitudes and strategies concerning phased retirement access from a survey among individuals who themselves are responsible for personnel policy at their organization. Conclusions are drawn in Section 5.

2. Literature review

A growing literature examines phased retirement's advantages and disadvantages for workers, firms, and the economy, including non-economic outcomes. For example, van Solinge et al.

(2023) show that in the Netherlands, phased retirement is associated with higher energy and lower fatigue, especially for workers with high work-related stress.

From a macro-economic perspective, an important issue is whether facilitating phased retirement leads to the desired increase in total hours of paid work. It may increase participation but also induce workers who would otherwise continue full-time to switch to part-time, so the net effect on total hours is unclear. Börsch-Supan et al. (2018) analyze policy reforms in many OECD countries that make retirement more flexible, including reforms that facilitate phased retirement. Considering male workers only, they find zero or negative effects on total hours worked, casting doubt on the efficacy of introducing more flexibility as a tool to stimulate old-age labor supply. The effect varies substantially with institutional context and the nature of the reform: negative effects on total hours are found for phased retirement programs in Austria (Graf et al., 2011) and Norway (Hermansen, 2015), whereas a 1996 German policy incentivizing phased retirement increased part-time work at the cost of retirement, raising total hours and generating net savings for the government (Berg et al., 2020). Huber et al. (2016) use linked employer–employee data from Germany to estimate the effects of firm-provided phased-retirement policies on male employees. They find that these policies reduce exits through unemployment and increase employment and earnings, although they may also induce a small share of workers to leave employment earlier under favorable labor-market conditions.

For the Netherlands, studies that analyze desired labor supply but do not consider possible restrictions from the demand side also find opposing results: Elsayed et al. (2018) and van Soest and Vonkova (2014) find that phased retirement reduces total desired hours worked, while Kantarcı et al. (2025) find a positive effect. If demand-side factors affect those who choose between phased and early retirement differently from those who choose between phased and late retirement, the effect on labor supply (i.e., on desired working hours) may differ from the effect on actual hours worked. This is one reason why studying employers' attitudes toward phased retirement is worthwhile.

Phased retirement can be understood as an age-related HR practice that redistributes employees' control over working time. Comparative evidence shows that such control is jointly shaped by national regulation, labor market conditions, and employer and union strategies, and may remain unevenly distributed across occupational groups (Berg et al., 2004). At the organizational level, firms choose HR practices considering production conditions and strategic considerations (Kaufman and Miller, 2011). Boehm et al. (2021) place these choices in a multilevel framework linking macro-level institutions, organizational HR policies, and their implementation for individual workers. Ollier-Malaterre et al. (2013) further show how compliance, peer-benchmarking, and business-case logics can produce heterogeneous organizational responses. Taken together, these perspectives suggest that institutions create opportunities for phased retirement, while organizations determine whether and how those opportunities are converted into actual access.

Clark and Ritter (2020a) see phased retirement as one of the responses of employers to the challenges of an aging workforce and a shortage of qualified workers, in addition to, for example, life-long training and development. Phased retirement allows older workers to remain on the job longer but also allows for some additional hiring. They emphasize, however, that phased retirement programs are popular only in specific sectors like higher education, where it is easy to organize part-time work or job sharing. Phased retirement programs in other organizations (including the federal US government) attracted only very few participants, often

due to lack of incentives for the workers or for the organization, or due to confusion about institutional arrangements like Social Security or Medicare. In line with this, Van Dalen (2021), showing that the use of partial pensions in the Netherlands is very limited and has even fallen since 2010, suggests that this is due to complexity and lack of employee incentives.

Clark et al. (2020b), based upon a survey among US employers in 2018, present the prevalence of the most common reasons why firms do not adopt formal phased retirement programs: “not aligned with workforce strategy” (42%), “difficult to administer” (32%), “prefer informal policy” (29%), “compliance challenges” (26%), “too costly” (15%), and “lack of employee interest” (11%). For the Netherlands, Henkens et al. (2021) conclude that both financial barriers and organizational barriers hamper employees taking phased retirement at a larger scale, while employers often consider the phased retirement decision as a private decision of the employee.

Turek et al. (2020) analyze data from two surveys conducted among employers in the Netherlands in 2009 and 2017. They consider several organizational policies aimed at managing an older work force. They treat part-time retirement as one of the “exit” policies and find that its use has fallen substantially between the two years, as well as the use of early retirement arrangements. On the other hand, over the same period, other policies such as ergonomic measures, training older workers, but also introducing more flexibility in working hours, have become much more popular.

The employer survey we use in this paper was strongly inspired by the work of Hutchens and Grace-Martin (2006) and Hutchens (2010), who analyze a survey on a representative sample of 950 US establishments conducted in 2001-2002, with questions that specifically focus on phased retirement of white-collar employees. Hutchens and Grace-Martin (2006) introduce a theoretical framework in which a minimum constraint on working hours of each employee can help the firm to obtain higher profits, depending on production technology. In line with the theory, they find that firms imposing minimum hours constraints and firms that are reluctant to offer part-time jobs in general, also tend to be less willing to allow phased retirement. This finding is confirmed by Albinowski (2024), using a quasi-panel covering 30 European Union countries for the period 2011 until 2021.

Hutchens and Grace-Martin (2006) also show that offering phased retirement is less common in smaller firms and in (other) firms that do not allow for job sharing or flexibility in terms of work schedules, again confirming that production technology matters. On the other hand, their empirical findings do not support the “pension hypothesis” that defined benefit pension systems would hamper phased retirement compared to defined contribution plans that typically offer more flexibility. Nor do they find support for the idea that firms would be more willing to offer phased retirement if employee groups that value phased retirement opportunities a lot are strongly represented (their “employee demand hypothesis”).

The same survey data set on US establishments was also used by Hutchens (2010), who focuses on the phased retirement opportunities of an individual white-collar employee with specific characteristics. After the generic question on access to phased retirement, the employer is asked to think of a specific older (55+) actual employee and to report how likely it is that this employee could shift into a part-time position. Hutchens finds that 83.3% of all establishments offer some form of phased retirement, but among those where phased retirement is feasible in principle, the specific selected employee has access to it in only 72.3%

of all cases. This already shows that employers are selective in whether they offer phased retirement to a specific employee or not.

Hutchens finds a strong positive relation between the specific worker's age and access to phased retirement. He gives two possible interpretations: Older workers are more likely to quit if they don't get phased retirement, or the older workers who are still there at an age close to 65 are the most productive and valuable ones that the firm really wants to keep. In line with the latter explanation, phased retirement is also more likely to be offered to "high performers" who work independently and require little supervision. Hutchens also finds that opportunities for phased retirement more often exist in jobs where part-time work is common and in jobs where it takes more effort to find a replacement. Other associations are not unambiguous, with the significance depending on which other variables are included in the specification. The evidence generally suggests that companies can use selection in phased retirement access as an incentive for older workers to encourage greater work effort. This is also consistent with the notion that most phased retirement arrangements are informal rather than formal generic agreements to which all workers of a specific age could be entitled.

Using similar survey data on Norwegian establishments, Hermansen and Midtsundstad (2015) find that 23% of companies offered a phased retirement program in 2010. These are typically the companies that also facilitate lifelong learning and take measures to prevent health problems, suggesting that such firms follow a holistic approach to active ageing.

3. Institutional setting and prevalence of phased retirement

The Dutch pension system

Retirement income in the Netherlands mainly stands on two pillars: the state pension and the occupational pension which is mandatory for more than 90% of all employees. The third pillar is private pension savings, but its share in retirement income is much smaller, except for the self-employed. The state pension scheme (AOW) pays a flat rate benefit to individuals who have reached the state pension age (SPA), independent of earnings, income or premiums paid. The benefit level depends on years of residence in the country and on household composition. For those who always resided in the Netherlands, it guarantees individuals older than the SPA a subsistence-level income. The scheme is unfunded and based on the pay-as-you-go principle: current state pensions are financed from the premiums paid by current workers (through the income tax). The SPA was fixed at age 65 for many years until 2013. Since then, it gradually increased to, e.g., age 67 for those born between March 1, 1957, and December 31, 1959. It will continue to rise further in the long run, with eight months for each additional year of increasing life expectancy. It does not allow for any form of flexibility in either the age of first claiming or the amount that is claimed.

An important difference between the Netherlands and the US or other Anglo-Saxon countries is that there is mandatory retirement at the SPA: Permanent contracts end by default at the SPA. Working beyond the SPA is possible if worker and employer agree, usually on a part-time and temporary basis with a new contract. Income taxation, social security entitlements, transition arrangements, etc. for post-SPA contracts differ substantially from the usual arrangements before SPA. In general, many employee rights negotiated in collective bargaining agreements or determined by law only apply before SPA and not for post-SPA contracts. On the other hand, social security premiums are lower after SPA, reducing the wedge between labor

costs and the net wage. All this makes post-SPA contracts cheaper and less risky and implies that the decision to offer phased retirement may depend on different factors for workers before and after their SPA.

Participation in the fully funded occupational pension scheme is mandatory for more than 90 percent of all employees. These pension schemes are organized in pension funds that are usually sector specific or are affiliated with one very large firm; employees cannot choose their own fund. The schemes are essentially individual but incorporate benefit arrangements for widow(er)s and orphans. The large majority are defined benefit pensions based upon average earnings over the whole career. Generous early retirement arrangements were prevalent until the early 2000's but were phased out since a tax reform in 2006 made them much less attractive. Today, many occupational pension funds allow maximum flexibility, allowing for early, late or partial retirement and pension claiming with actuarially fair trade-offs and a fair price for retiring early or working longer. In response, the average retirement age increased from 61 in the early 2000's to more than 66 in 2024.¹

Part-time work and partial retirement

In all age groups, part-time work in the Netherlands is widespread among women as well as men, although not to the same extent. According to Eurostat, 60.5% of employed women worked part-time compared to 18.6% of employed men in 2024.² These high rates of part-time work among men and women are among the highest in Europe. In comparison, according to Eurostat, the average percentage of part-time workers in the EU is approximately 27.7% for women and 7.7% for men.

Dutch employers are normally obliged to honor requests of their employees to reduce working hours unless there are good reasons (e.g., production technology) not to. Particularly in the context of phased retirement for older workers, this is rooted in a comprehensive legislative and policy framework aimed at promoting work-life balance, labor market flexibility, and the well-being of the aging workforce. This framework encompasses labor laws, collective agreements, and pension policies that collectively support the gradual transition of older employees from full-time work to (full) retirement. Collective labor agreements (CAOs) typically concern the sector level and apply to unionized as well as non-unionized workers; the unionization rate of an individual firm usually does not play a large role.

The comprehensive Dutch legislative framework supports work-life balance and labor market flexibility, providing a foundational right for employees to adjust their labor supply. Historically rooted in the Working Hours Adjustment Act since 2000, this legal structure was significantly expanded by its 2016 successor, the Flexible Working Act. Under this statutory framework, employees with an organizational tenure of at least 26 weeks have the right to formally request changes to their working hours, schedule, or place of work. Employers are legally required to honor employee requests for reduced hours unless they can demonstrate compelling, substantial business interests for refusal. This statutory right is particularly significant for older workers seeking phased retirement, as it facilitates a gradual reduction in work intensity during their career-ending years. In line with these institutional options to

¹ Source: [Gemiddelde pensioenleeftijd voor het eerst boven de 66 jaar | CBS](#)

² Source: https://ec.europa.eu/eurostat/databrowser/view/lfsi_pt_a_custom_22301284/default/table

downscale hours within the same firm, phased retirement is more prevalent in the Netherlands than bridge jobs as a form of partial retirement (Bloemen et al., 2015).

Most occupational pension schemes accommodate part-time pensions, so that employees can choose to partially retire and start receiving a portion of their occupational pension benefits while continuing to work part-time. This differs from the completely inflexible state pension.

The rising SPA and the phasing out of generous early retirement schemes hampered early retirement for many older workers, including those with health issues or demanding occupations. In response, since the late 2010's, CAOs between employers or employer organizations and unions included provisions for reducing working hours, flexible work arrangements, and phased retirement. Many CAOs include seniority schemes that provide older employees with the option to work reduced hours with a less than proportional reduction in earnings. These schemes often kick in at a certain age (like five years before SPA) and are designed to accommodate older workers who have difficulties working full-time until the higher SPA, but also to retain the expertise of senior employees. The arrangements allow employees to work fewer hours in the final years before their SPA with a less than proportional reduction in earnings and full pension accruals (see, e.g., Rutten et al., 2025). Details vary across sectors; an example is the 60/70/100 arrangement: work 40% less, receive 70% of the (gross) wage, and accumulate 100% of occupational pension rights.

Phased retirement prevalence and intentions

We document to what extent phased retirement is relevant to employees and whether it involves a change in type of work, employer, sector and hourly wage using a survey focused on phased retirement preferences and opportunities from June 2017 (unfortunately, more recent data are not available). The survey was fielded to participants of the LISS panel, an ongoing panel study on a large representative sample of the Dutch non-institutionalized population of ages 16 and older (see Scherpenzeel, 2011). The panel is based upon a random probability sample of Dutch residents drawn by Statistics Netherlands. It is administered over the Internet but also covers households that do not have Internet access, providing them with the necessary equipment. Households in the panels answer survey questions every month, on a variety of topics. There are annual core questionnaires, but the panel also allows academic researchers to field their own surveys. The specific survey covered individuals of age 35 and older (assuming younger individuals have not yet thought about retirement). Our final sample has 3,999 respondents.

In this specific survey we collected information on expected (for those who did not yet retire) or realized (for those who retired) retirement behavior. We asked respondents to construct a sequence that corresponds as much as possible to their actual behavior or their current plans for the remainder of their labor market career. For each two years age category 55-56, ..., 67-68, 69-plus, they indicated their main labor market status, choosing among full-time work, part-time work, or (fully) retired. They then also indicated, at each age, (expected) changes in the type of work, employer, the sector, and the hourly wage. See Figures A1 and A2 in the Appendix for the exact questions.

Table A1 in the Appendix shows the most reported sequences. The fraction of respondents who report a sequence that indicates gradual retirement – full-time work succeeded by part-time work and then full retirement – is 26.3%. This fraction falls to 14.6% if we only consider respondents who have retired, so that the sequence is not shaped by expectations but only by

actual decisions. This difference may be a cohort effect or time trend but also may mean that people often do not follow up on what they expect, possibly due to demand side restrictions. Among all phased retirement sequences, only 15.2% involve a change in type of work, 8.1% a change in employer, and 6.9% a change in sector. These figures suggest that in most cases gradual retirement means phased retirement in the career job. In 22.5% of the cases, however, gradual retirement involves a reduction in hourly wage. This is in line with several US studies that found that employees face a part-time wage penalty when they switch to a less demanding job in a phased retirement arrangement (Hutchens, 2010; Ameriks et al., 2020). Finally, distinguishing between men and women, we find that 37.9% of men (expect to) participate in gradual retirement versus 14.4% for women. This may be because women much more often work part-time during their career and few part-timers are interested in phased retirement, but it could also be because women do not perceive phased retirement as a feasible option. Whether that would be a justified perception is one of the issues we investigate in the next section.

4. Employers' views on phased retirement

Following Hutchens and Grace-Martin (2006), we investigate the organizational generic policy concerning phased retirement, but, along the lines of Hutchens (2010), we also analyze the likelihood that a specific worker will be offered phased retirement. The theoretical framework of Hutchens and Grace-Martin identifies several factors that may drive these decisions; we briefly summarize their main conclusions, based upon the existing literature. First, the type of work done in the organization matters for the generic policy: organizations in which part-time work is common are more likely to offer phased retirement, since creating part-time jobs for older workers will be easier (or less costly) to organize. How easy it is to accommodate part-time work may also depend on the sector of industry and the size of the organization.

The second factor that may play a role is workers' demand, suggesting that phased retirement may be more likely in organizations with many older workers and few younger workers, and organizations with a large share of female workers – Hutchens and Dentinger (2003, p. 263) find that women are more interested in phased retirement than men are, though this may not apply in the Netherlands where the majority of women work part-time already.

The third factor is related to the type of pension arrangement: Phased retirement may be unattractive in a defined benefit context. Hutchens and Grace-Martin find no evidence that this matters and we do not consider this in our empirical analysis – most of the workers have a defined benefit arrangement that allows for phased retirement in an actuarially neutral way.

For a specific worker with a specific job, the nature of the job will matter in a similar way for the generic policy of the firm: Offering phased retirement is more likely for workers in jobs where flexibility in working hours or splitting the work into different jobs is easier or less costly. Organizing part-time work may also be easier in jobs where someone mainly works alone rather than in a team. It may also depend on how difficult it is to find a replacement for the specific tasks the job involves. Whether phased retirement is offered to a specific worker may also depend on the characteristics of that specific worker. For example, the employer may not want to run the risk of losing workers that perform very well, have acquired job specific skills through extensive on the job training, or need little supervision.

Data

In May 2024, a survey was conducted in the LISS panel on employers' preferences for phased retirement. LISS panel members who indicated (in February 2024) that they have a management position in their organization were invited to take part in the survey. 37.1% of them reported they were themselves responsible or co-responsible for personnel policy in their organization, for example for hiring new people (e.g. as an employer or personnel manager). The other managers were asked for their perceptions of how decisions about personnel policy are made in their organization. Moreover, employers who are members of a large Dutch employer organization (AWVN) or the association of Dutch municipalities (the employers of civil servants; VNG) were invited to participate in the same survey through AWVN or VNG. Employers were asked to find a person in their organization with the right expertise to answer the survey questions on the organization's behalf. 72.6% of the respondents were themselves (co-) responsible for personnel policy. These observations were added to the sample, giving it a total (gross) sample size of 799. The sample underrepresents small firms and overrepresents those in the secondary and tertiary sectors. We use entropy balancing (Hainmueller, 2012) to create weights to account for this: observations are reweighed to match the firm-size and sector distributions in the population reported by Statistics Netherlands.

Following the survey design of Hutchens and Grace-Martin (2006) and Hutchens (2010), the survey consisted of three parts. The first part had questions about establishment characteristics and personnel policy in general. The second part was specifically about the generic personnel policy for employees aged 55 or older who might want to move from full-time to part-time work. These questions are very similar to the questions analyzed by Hutchens and Grace-Martin (2006). The third part asked respondents to think about a specific employee (randomly selected among three older employees in the organization) and their background, job characteristics and perceived opportunities for and restrictions on phased retirement. These questions were the focus of Hutchens (2010).

Table 1 presents the four variables of main interest for analyzing organizational attitudes and strategies concerning phased retirement opportunities in general (panels a and b) and for a specific worker (panels c and d).³ More than 93% of the respondents report that their organization would honor requests for phased retirement (panel a). In most cases this applies to every request but in 17% only in specific cases. On the other hand, most organizations do not have a formal written policy for a transition from full-time to part-time work (panel b). And if there is such a policy, it is often applied with exceptions. The answers to the two questions in panels a and b are strongly correlated (not shown in the table): access to phased retirement is much more common in organizations where a formal policy for requesting a shift to part-time work is present.

A specific older worker (randomly selected from at most three workers approaching retirement mentioned by the respondent) is often likely to get access to phased retirement, particularly if the worker has passed the state pension age (SPA) (panels c and d of Table 1). This

³ The number of observations is smaller than 799 due to item non-response ("don't know" answers) and the routing of the questionnaire. For example, questions related to phased retirement were only asked if there were workers older than 55 in the organization. Moreover, Questions 2 to 4 in Table 1 were only asked if the answer to the first question was "yes" or "in some cases", if the respondent knew at least one worker older than 55 of given gender in the organization, and if the respondent indicated at an earlier question that the organization in principle could allow phased retirement before (for panel c.) or after (for panel d.) SPA.

is in line with the notion that working after SPA is not very common (and the large percentage “very unlikely” probably concerns organizations where the organization does not do this at all), but if workers are employed after the SPA, it is often on a part-time (and temporary basis).

Table 1. Employer attitudes towards phased retirement

<i>a. Consider an employee aged 55 or older with a permanent full-time job. This employee expresses the wish one day to want to work less at some time in the coming years. Would your firm honor this request?</i>	
	Percentage
No	6.61
In some cases yes, in others not	17.27
Yes	76.12
N	648
<i>b. Which of the following statements best expresses your firm's policy concerning shifts from full-time to part-time schedules?</i>	
	Percentage
Decisions are made informally on a case-by-case basis	74.62
A formal written policy is in place, but exceptions can be made in individual cases	9.14
A formal written policy is in place that applies to all employees	16.24
N	417
<i>c. How likely is it that the selected worker could switch to a part-time position before the state pension age?</i>	
	Percentage
1 Not at all likely	9.40
2	21.03
3	12.37
4	28.24
5 Very likely	28.96
N	373
<i>d. How likely is it that the selected worker could switch to a part-time position after the state pension age?</i>	
	Percentage
1 Not at all likely	13.07
2	2.36
3	12.84
4	33.90
5 Very likely	37.82
N	308

Notes: Percentages are adjusted using entropy balancing weights ensuring the sample is representative across firm sizes and industry sectors.

The explanatory variables, motivated by the conceptual framework sketched above, are defined and summarized in Tables A2 and A3. The first set in Table A2 are general characteristics of the organization possibly associated with how easy it is to organize phased retirement, followed by variables characterize the organization's work force. The variables "flexible work times" and "job sharing" indicate the organizational attitude towards non-standard work schedules. The variables in Table A3 refer to characteristics of the specific worker that the last two questions in Table 1 refer to (age, gender, education; indicators of the worker's job performance; characteristics of the job of the selected worker). These variables are obviously missing if the respondent could not identify specific workers older than 55 of given gender.

Regression results – organizational strategy

We first investigate the answer to the question whether the organization would offer phased retirement to a generic employee (panel a in Table 1). A multinomial probit model is estimated that sets out whether the answer to the question is "no", "in some cases", or "yes". The estimated average marginal effects of the explanatory variables on each of the three possible outcomes are presented in Table 2.

The association between offering phased retirement and firm size is positive, though not statistically significant. Small firms lack HR policies in general and may not so easily be able to divide tasks efficiently when people fall out, whether partially or fully. The positive association is consistent with the literature review of Hermansen and Midsundstad (2015), who found that firm size is positively associated with offering retention measures and phased retirement schemes. Their explanation is that larger companies have greater HR capacity and resources to support these policies. It is also in line with Remery et al. (2003) who conclude that organizational friction and staffing constraints in smaller firms are one of the main explanations for limited access to part-time work or phased retirement in the Netherlands.

We find no significant association with the dummy "not for profit". On the one hand, non-profit organizations might offer phased retirement more easily than for-profit firms, as they often give high priority to worker-oriented HR policies and employee well-being. On the other hand, non-profit organizations often operate under strict funding limits and lean staffing, which could make it harder to absorb the extra costs of part-time workers.

Somewhat surprisingly, organizations in the services sector are significantly less likely to offer phased retirement than the primary sector. One possible explanation is that the discussion on allowing for phased retirement centers around physically demanding occupations where working until SPA is difficult, and such occupations are less common in the services sector. Another explanation could be that those who are potentially interested in phased retirement already work part-time at an earlier age.

In line with earlier empirical and theoretical work, the probability of offering phased retirement is significantly higher in firms with a larger proportion of white-collar jobs. Grounding their model in employer coordination costs, Hutchens and Grace-Martin (2006) explain that white-collar positions generally carry lower team interdependence and lower fixed administrative setup costs, making work-hour reductions easier for employers than in blue-collar manufacturing or shift-based setups. Similarly, Allen, Clark, and Ghent (2004) frame phased retirement through transaction costs and team coordination, observing that phased retirement is predominantly extended to white-collar, faculty, and administrative staff because their tasks are inherently modular and not bound to rigid team production schedules.

The presence of many part-time workers in a firm is associated with a higher likelihood of offering phased retirement schemes, and this association is marginally significant. This positive relation is in line with the theoretical arguments and the findings of Hutchens and Grace-Martin (2006). The percentage of female workers and the percentage of workers older than 55 do not play a significant role.

Organizations that have a large percentage of shift workers seem to offer phased retirement selectively – they often offer phased retirement in some cases but significantly less in all cases. This might be because part-time work and shift work are hard to schedule at the same time. Organizations that allow for flexible working times much less often offer phased retirement in some cases only; instead, they often offer phased retirement as a general arrangement. Finally, as expected, offering job sharing is positively connected to offering phased retirement. This is in line with the argument that phased retirement is easier in organizations that already work with part-time arrangements.

Table 2. Multinomial probit model explaining whether the employer allows phased retirement: Average marginal effects on the probability of each outcome

	No	In some cases yes, in others not	Yes
<i>Establishment characteristics</i>			
Establishment size category	-0.037 (0.027)	0.018 (0.031)	0.019 (0.038)
Organization is non-profit	0.018 (0.052)	-0.138 (0.110)	0.120 (0.113)
Sector is secondary	0.144* (0.081)	0.036 (0.123)	-0.180 (0.138)
Sector is tertiary	0.220*** (0.080)	0.066 (0.113)	-0.286** (0.126)
Sector is quaternary	0.105 (0.078)	0.044 (0.133)	-0.149 (0.141)
<i>Workforce composition</i>			
White-collar employment-share category	-0.043*** (0.013)	0.025*** (0.009)	0.017 (0.013)
Share of employees aged 55 or older, category	0.003 (0.007)	-0.002 (0.013)	-0.002 (0.014)
Female employment-share category	-0.001 (0.007)	-0.010 (0.015)	0.011 (0.017)
Shift-work share category	0.006 (0.005)	0.026* (0.014)	-0.032** (0.015)
<i>Proxies for minimum hours constraints</i>			
Share of employees working fewer than 35 hours, category	-0.016** (0.008)	-0.007 (0.012)	0.023* (0.013)
Flexible working times	0.085* (0.051)	-0.234*** (0.075)	0.150* (0.084)

Job sharing	-0.151*** (0.049)	0.038 (0.069)	0.113 (0.078)
N	511		
Log pseudolikelihood	-228.530		
Pseudo R-squared	0.339		
p-value of Wald test for model significance	0.000		

Notes: 1. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$. 2. Standard errors based on the delta method. 3. Marginal effects may not sum exactly to zero due to rounding. 4. Establishment size and workforce-share variables are based on ordered survey response categories and are entered as numerical category codes; see Table A2. Their average marginal effects therefore refer to moving up one response category. Binary variables are coded as indicated in the table.

Table 3 presents the multinomial probit marginal effects for the second question in Table 1, indicating whether the firm has a policy concerning requests for shift to part-time work. The base outcome is that there is no such policy; if there is a policy, it can be fully generic or allow for exceptions. As expected, the probability that there is a general policy increases with the size of the organization, since larger firms generally have more formal HR policies. In non-profit organizations, a policy that can be applied with exceptions is relatively common. This is less common in the services sector. In the secondary sector (industry, construction, etc.), it is more likely that there is no formal policy rather than a policy that applies without exception.

The variables that refer to the composition of the workforce of the organization play a larger role in Table 3 than in Table 2. Organizations with many older workers more often have a policy that applies to everyone than a policy that allows for exceptions. When a significant portion of the workforce approaches retirement, case-by-case discretionary arrangements become operationally inefficient and raise concerns over procedural fairness; standardized policies provide essential clarity and equity. Surprisingly, we find a negative relation between having a generic policy and the fraction of part-time workers. As argued before, this finding may reflect the redundancy of formal phased retirement in flexible work environments: If part-time employment is already the norm, older workers can downscale their hours through standard contractual adjustments and age-specific phased retirement schemes are not needed.

Organizations with a large fraction of female workers significantly less often have a policy that allows for exceptions. Arguably this could be due to the fact that organizations with a higher proportion of female employees are concentrated in sectors with CAOs that prioritize formal, institutionalized rules over managerial discretion.

Table 3. Multinomial probit model for organization's policy concerning shifts to part-time work: Average marginal effects on the probability of each outcome

	Decisions are made informally on a case-by-case basis	There is a formal written policy, but adjustments can be made	There is a formal written policy that applies to everyone
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		in individual cases	
<i>Establishment characteristics</i>			
Establishment size category	-0.119*** (0.032)	0.020 (0.018)	0.100*** (0.022)
Organization is non-profit	-0.406*** (0.124)	0.316*** (0.109)	0.090 (0.081)
Sector is secondary	0.313** (0.141)	0.002 (0.081)	-0.314** (0.128)
Sector is tertiary	0.291** (0.120)	-0.138* (0.074)	-0.153 (0.107)
Sector is quaternary	-0.048 (0.138)	-0.061 (0.076)	0.109 (0.116)
<i>Workforce composition</i>			
White-collar employment-share category	0.025** (0.010)	-0.012* (0.006)	-0.013* (0.007)
Share of employees aged 55 or older, category	0.002 (0.018)	-0.038** (0.018)	0.036*** (0.009)
Female employment-share category	0.014 (0.017)	-0.036*** (0.012)	0.022* (0.013)
Shift-work share category	0.013 (0.014)	-0.003 (0.007)	-0.010 (0.012)
<i>Proxies for minimum hours constraints</i>			
Share of employees working fewer than 35 hours, category	0.044*** (0.014)	-0.006 (0.009)	-0.038*** (0.011)
Flexible working times	-0.151* (0.080)	0.123*** (0.045)	0.028 (0.076)
Job sharing	0.173** (0.069)	-0.073* (0.038)	-0.100 (0.064)
N	357		
Log pseudolikelihood	-124.199		
Pseudo R-squared	0.467		
p-value of Wald test for model significance	0.000		

Notes: 1. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$. 2. Standard errors based on the delta method. 3. Marginal effects may not sum exactly to zero due to rounding. 4. Establishment size and workforce-share variables are based on ordered survey response categories and are entered as numerical category codes; see Table A2. Their average marginal effects therefore refer to moving up one response category. Binary variables are coded as indicated in the table.

Regression results – selected worker

Table 4. Ordered probit models explaining the likelihood that the selected worker can switch to a part-time position before or after the state pension age

	Likelihood that the selected worker could switch to a part-time position <i>before</i> the state pension age	Likelihood that the selected worker could switch to a part-time position <i>after</i> the state pension age
<i>Demographic characteristics of selected worker</i>		
Age	-0.077* (0.045)	0.196** (0.087)
Education is higher	0.038 (0.133)	0.701*** (0.245)
Male	1.108** (0.501)	1.444** (0.689)
<i>Job performance of selected worker</i>		
Learns new procedures and technologies	0.414 (0.259)	0.396 (0.299)
Creative in solving problems	0.163 (0.292)	-0.361 (0.264)
Willing to make extra effort to get job done	-0.735*** (0.239)	-0.358 (0.326)
Gets along well with coworkers	-0.285 (0.190)	0.108 (0.259)
Has energy level needed to perform well in the job	0.087 (0.270)	0.483* (0.258)
<i>Occupational category of selected worker</i>		
Intermediate non-manual	2.220*** (0.469)	2.518*** (0.634)
Skilled manual or technical; low-skill or basic	3.249*** (0.587)	3.518*** (0.967)
<i>Job characteristics of selected worker</i>		
<i>Job is compatible with part-time work</i>		
There are regular part-timers in selected worker's job title	0.233*** (0.079)	0.196* (0.108)
Job involves tasks that are discrete	0.313* (0.182)	0.606*** (0.204)
Most tasks are performed alone	0.075 (0.190)	-0.458** (0.190)
Job requires close communication with others	-0.405 (0.332)	-0.479* (0.289)
<i>Skill and training</i>		

Jobs where old are more productive than young	0.467** (0.237)	0.719*** (0.260)
Selected worker's job tenure in establishment	-0.466*** (0.153)	-0.032 (0.214)
Job requires several years of experience to do well	0.093 (0.328)	0.755** (0.353)
Employer-sponsored training	-0.722*** (0.206)	-0.385* (0.230)
<i>Difficulty in finding a replacement</i>		
Desirable to fill the job of worker with a recent graduate	-0.571** (0.268)	-0.062 (0.267)
Increased difficulty finding employees for selected worker's job type	0.909** (0.374)	-0.129 (0.414)
N	220	180
Log pseudolikelihood	-184.982	-142.623
Pseudo R-squared	0.324	0.337
p-value of Wald test for model significance	0.000	0.000

Notes: 1. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$. 2. Standard errors based on the delta method. 3. The outcome variable is assessed using a 5-point Likert scale ranging from 1 ("Not at all likely") to 5 ("Very likely"). 4. Base occupational category: High-skill non-manual.

In Tables 2 and 3, we focused on how firm characteristics are associated with phased retirement arrangements, without specifying the characteristics of the worker. Since Hutchens (2010) shows that employers can be selective in offering phased retirement, we now consider the case in which the employer needs to decide on whether to offer phased retirement to a specific worker, explaining the answers to questions c and d in Table 1. Following Hutchens (2010), we estimate which observed worker characteristics (as perceived by the employer) are important for the employer's decision to offer phased retirement or not. Due to the important role of the state pension age (SPA) in the Dutch institutional context, we separately consider phased retirement before (Question c in Table 1) and after (Question d) SPA. As we discussed in Section 3, most employees face mandatory retirement at the SPA. Phased retirement before SPA can be an alternative for abrupt retirement at SPA but it can also be an alternative for early (full) retirement. Full-time work after SPA is rare, implying that phased retirement after SPA is almost always an alternative for not working at all. Moreover, the costs are different: the wedge between gross and net wage is much lower after retirement, since social security premiums (mainly the state pension premium) do not apply after SPA and the employer has fewer obligations (compensation, help with finding another job, sickness benefits, etc.) if the contract is terminated after rather than before SPA.

We use ordered probit models to explain the likelihood of a phased retirement offer from worker characteristics. The results are presented in Table 4. Since the dependent variable takes value from 1 ("not at all likely") to 5 ("very likely"), a positive coefficient means that an increase

in the explanatory variable raises the likelihood of a phased retirement offer, keeping other variables constant.⁴

Older workers are more likely to be offered phased retirement after SPA, in line with Hutchens (2010). This could be a selection effect if workers stay longer at the organization if they perform better, making it more attractive to keep them after their SPA. Before SPA, the age effect is smaller and less significant. A similar difference is found for education: it is not related to offering phased retirement before SPA but has a strong and significant positive association with access to phased retirement after SPA, suggesting that higher education makes the worker more valuable for the organization. Higher educated workers are probably also kept in order to transfer useful skills to their younger (new) colleagues.

An unexpected result is that, keeping all the other variables constant, male workers are much more likely to get access to phased retirement than female workers do, both before and after SPA. The average *ceteris paribus* difference in the probability that phased retirement is offered is more than 20 percentage points before SPA and more than 30 percentage points after SPA. Note that the question considers full-time workers, so the fact that many women work part-time cannot be the explanation. Formal arrangements and collective bargaining agreements do not make any distinction between male and female workers, and we control for occupational category and job and worker characteristics, making this result particularly surprising. With the current sample size, it seems impossible to pin down the reason for this finding, but it seems an interesting topic for follow-up research.

The coefficients on the occupational level show that phased retirement is unlikely for managers and professionals (high skilled, nonmanual – the omitted category).⁵ This is probably the type of job for which part-time work is difficult to organize, and the cost of splitting tasks is large. In line with the literature, we find that workers in a position where part-time work is more common are more likely to gain access to phased retirement, particularly prior to SPA. This is also in line with our earlier results for a generic worker, although the relation seems weaker here. As expected, phased retirement is more often offered in jobs where older workers are more productive than younger workers, both before and after SPA. Other job characteristics do not do much for phased retirement before SPA but do seem to matter after SPA. Jobs that require a lot of experience make a phased retirement option after SPA more likely, probably because for such jobs switching to another worker is costly and the older part-time workers can help the replacement worker to build up the required skills.

The variables that directly refer to finding a replacement are significantly associated with phased retirement opportunities before SPA only. If it is desirable to find a replacement that recently graduated, this reduces the probability to offer phased retirement, whereas increasing

⁴ The estimated average marginal effects of the explanatory variables across the five ordered outcomes are presented in Tables A4a and A4b.

⁵ We also re-estimated both models controlling for organization size, non-profit status, and industry sector. Model fit improves (pseudo R-squared rises from 0.32 to 0.39 for switching to part-time before state pension age and from 0.34 to 0.47 for switching after), and non-profit status emerges as a significant positive predictor in both models. Most original significant coefficients remain significant with similar or stronger magnitudes, though several variables — including worker's age, tenure, job autonomy, desirability for recent graduates, and physical stamina requirements — become more statistically significant after accounting for these organizational characteristics. This suggests our main findings are robust, but the baseline specifications may have understated some effects due to omitted organizational factors.

difficulty in finding a replacement is positively related to offering phased retirement. For the latter, the average marginal effect on the probability that a phased retirement offer is very likely, is substantial: 14 percentage points.

5. Discussion and conclusions

This study examines how Dutch organizations translate institutional possibilities for phased retirement into access for individual workers. Most organizations offer some form of phased retirement, but formal policies covering broad employee groups are considerably less common than case-by-case arrangements. The absence of a formal policy is strongly negatively associated with phased-retirement opportunities. Although our cross-sectional analysis cannot establish causality, the results suggest that institutional possibilities are more effective when organizations incorporate them into explicit HR policies.

Access also depends on work organization and worker characteristics of the selected worker. The prevalence of part-time work in the occupation, older workers' productivity relative to younger workers, and replacement difficulties are all associated with phased-retirement opportunities. Replacement constraints matter more before the state pension age (SPA), whereas education and experience are more strongly associated with access after SPA. We also find that, conditional on occupation, job, and worker characteristics, male full-time employees are substantially more likely than comparable female employees to be offered phased retirement. This unexpected association warrants further research and raises concerns about the distributional consequences of case-by-case discretion.

These findings fit a multilevel institutional and organizational perspective. Comparative research shows that employee control over working time is jointly shaped by national regulation, labor-market conditions, and employer and union strategies, and may remain unevenly distributed across occupations. At the organizational level, firms choose HR practices in light of production conditions and strategic considerations. Age-related HR practices can be placed in a framework connecting macro-level institutions, organizational HR policies, and their implementation for individual workers. Organizations may respond differently to similar institutional conditions depending on whether compliance, peer benchmarking, or the perceived business case dominates decision-making. The Dutch case therefore illustrates a more general point that likely applies beyond the Netherlands: institutions create opportunities for phased retirement, but organizations determine whether and how those opportunities are converted into actual access.

Our main policy recommendation is therefore not simply to expand the legal or financial availability of phased retirement. Policy should also help organizations translate existing opportunities into transparent and workable HR practices. Formalization does not require an inflexible entitlement that applies identically to every occupation. Our results indicate that genuine differences in work organization and replacement possibilities matter. Sectoral or organizational policies could instead establish default eligibility conditions, a clear application procedure, transparent grounds for exceptions, and opportunities for employees to request reconsideration. Such arrangements would preserve room for operational constraints while reducing the uncertainty and potential inequality associated with entirely informal decisions.

In the Netherlands, collective labor agreements provide an important vehicle for such formalization. Subsidized partial-retirement arrangements such as the Generation Pact

illustrate how employers, employees, and pension providers can jointly address reductions in working hours, earnings, and pension accrual. Their broader relevance lies in this coordination mechanism rather than in the details of the Dutch scheme. In other institutional settings, sectoral bargaining bodies, employer associations, or public agencies could perform a similar role. Expansion of these arrangements should, however, be accompanied by evaluation of participation, employment duration, total hours worked, employer costs, and the distribution of access across occupations and worker groups.

Equitable access should be an explicit objective. Case-by-case arrangements allow employers to retain workers whose knowledge, productivity, or experience is particularly valuable, but they may also make access dependent on managerial assessments and the ease with which particular jobs can be reorganized. Workers in occupations in which part-time work is difficult to arrange may consequently have less access, even when reducing working hours would help them remain employed. The gender difference in our estimates provides an additional reason for caution. Because we cannot determine whether this association reflects unobserved worker or job characteristics, negotiation processes, or managerial discretion, it should not be interpreted as causal evidence of discrimination. Nevertheless, organizations and social partners could monitor applications and decisions by gender, occupation, earnings, and other relevant characteristics. Transparent criteria and periodic reporting would make it easier to identify whether formally neutral arrangements produce systematically unequal outcomes.

Because post-SPA employment carries lower regulatory costs and liability, employers are more open to part-time arrangements after mandatory contract termination. Post-SPA phased retirement is granted mainly based on selective worker traits, retaining highly educated and experienced talent. Before SPA, organizational constraints, such as replacement difficulty and part-time availability, matter more than individual education or experience. Policy interventions should therefore target pre-SPA institutional barriers. Aligning pre-SPA flexible work regulations with the low-cost, adaptable structure of post-SPA arrangements can incentivize employers to offer gradual retirement during core working years, helping more workers adjust workloads before the statutory retirement age. More generally, the Dutch case suggests that discontinuities in employment protection, pension eligibility, and contract rules around the retirement age can shape both the prevalence and the selectivity of phased retirement.

Employer awareness and learning remain relevant, but information campaigns alone are unlikely to be sufficient. The institutional-logics perspective suggests that organizations may respond to regulatory requirements, practices adopted by comparable employers, or evidence that phased retirement serves a strategic business interest. Governments, social partners, and employer associations can address all three channels by clarifying legal requirements, disseminating comparable sectoral practices, and documenting the costs and benefits of retaining older workers. Practical guidance should include examples of job redesign, knowledge-transfer arrangements, replacement strategies, and transparent decision procedures. Where employer costs are demonstrably an important obstacle, targeted and temporary subsidies may be appropriate, but they should be linked to clear objectives and evaluated, rather than offered as general incentives.

Flexible pension arrangements can complement workplace flexibility by allowing employees to reduce their working hours without an equally large decline in current income. Conversely, pension rules can constitute a barrier to flexible and part-time work among older employees (Loretto et al., 2005). Partial pension withdrawal is already possible within much of the Dutch

occupational pension system, but implementation and communication vary across sectors and organizations. Pension providers could make these options more accessible by showing workers the combined consequences of partial pension income, part-time earnings, pension accrual, and eventual full-retirement income. They can also support sector- or firm-specific programs by coordinating pension rules with collective agreements and organizational HR policies. The transferable lesson is that pension flexibility and workplace flexibility must be designed jointly: a partial pension is of limited value if the employer does not permit a corresponding reduction in working hours.

Finally, the appropriate objective is not necessarily to maximize participation in phased retirement. Existing studies reach different conclusions about its effects on total labor supply and public finances. Firm-provided phased-retirement policies can increase employment and earnings and reduce exits through unemployment, but they may also encourage earlier labor-market exit among some workers. Evaluations should therefore look beyond take-up and consider effects on employment duration, total hours worked, hiring to replace reduced hours, employee well-being, employer costs, and the distribution of access across worker groups.

The central policy lesson is that creating a formal possibility for phased retirement is only a first step. Effective policy must also address how organizations implement that possibility, how jobs are adapted to reduced working hours, and how access is distributed among workers. Combining pension and employment institutions that enable flexible work with transparent organizational policies, flexible job design, and systematic evaluation offers the most promising route toward phased retirement that extends working lives without reinforcing inequalities in access.

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Online Appendix

Questions about labor market status at each age (LISS panel 2017; see Section 3)

People follow different employment paths. Some work full-time and then fully retire, some retire only partially, others do something else. Below we show a timeline that intends to outline your employment path from age 55 and onwards. The line indicates eight age categories. Below each category is a drop-down menu which lists four work status choices. Please choose your **past and expected future** work status at all given ages.

- Choose 'Full-time' for 35 or more hours of work per week.
- Choose 'Part-time' for less than 35 hours of work per week.
- Choose 'Fully retired' for full retirement.
- Choose 'Other' if you are unemployed, sick, on other leave, or homemaker.
- If you have/had more than one work status during a given period, choose the one in which you spent most of your time.
- When choosing a future work status, take account of what your **opportunities** will allow you to do. For example, if your employer prohibits part-time work, avoid choosing part-time from the list.

	Age 55-56	Age 57-58	Age 59-60	Age 61-62	Age 63-64	Age 65-66	Age 67-68	Age 69+
Work status	Full-time	Full-time	Full-time	Part-time	Part-time	Part-time	Retired	Retired

Figure A1: Question asking to outline past and expected future work status from age 55 onwards

The employment path below outlines your work status from age 55 and onwards. Below each age category and at the corresponding work status, we present drop-down menus where we would like you to indicate **past and expected future changes** in your type of work, employer, or industry, or increase or decreases in your hourly wage.

Please note that **you only have to click for changes**. For example, if you change your employer twice, say at ages 55 and 61, then choose 'Change' at the corresponding row 'Employer' and age categories '55-56' and '61-62'. If there is no change, do not make any selection in the menus.

	Age 55-56	Age 57-58	Age 59-60	Age 61-62	Age 63-64	Age 65-66	Age 67-68	Age 69+
Work status	Full-time	Full-time	Full-time	Part-time	Part-time	Part-time	Retired	Retired
Type of work			Change					
Employer			Change					
Sector			Change					

Hourly wage			Increase					
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Figure A2: Question asking to indicate (expected) changes at respective ages in the type of work, employer and the sector in which the respondent work(ed) and increases and decreases in hourly wage.

Table A1: Most common self-reported retirement sequences

Sequence	Percent	Sequence	Percent
22222333	6.50	11222333	1.08
22222233	6.25	11123333	1.05
44444444	5.15	13333333	1.05
11111133	4.08	22333333	0.95
11111333	3.90	11223333	0.93
44444333	3.78	11122223	0.83
11113333	3.75	11222233	0.83
22223333	3.68	11112223	0.78
22233333	3.25	11111122	0.73
11133333	3.08	11111222	0.68
11122333	2.80	23333333	0.68
11112233	2.70	11111112	0.65
22222223	2.60	12223333	0.63
33333333	2.58	11144333	0.60
44444433	2.40	11444333	0.60
11111233	2.20	22244333	0.58
11122233	2.18	12222333	0.53
11112333	1.80	12233333	0.40
11111113	1.73	22444333	0.40
11111123	1.65	11233333	0.35
11111223	1.63	14444333	0.35
11333333	1.55	44444443	0.35
22222222	1.30	12222233	0.33
11111111	1.08	12333333	0.33

Notes: 1. 1: Full-time work, 2: Part-time work, 3: Retired; 4: Other. 2. Retirement sequences are ranked according to the percentage of 3,176 respondents who reported the sequence. 3. The eight elements of a given sequence refer to the self-reported work status at eight age categories, given by 55-56, 57-58, 59-60, 61-62, 63-64, 65-66, 67-68, and 69 plus.

Table A2: Explanatory variables at the level of the organization

Question	Response options	Mean	Std. dev.
<i>Establishment characteristics</i> About how many people are employed at your organization? If you do not know, give your best estimate.	1. 1–25; 2. 26–50; 3. 51–100; 4. 101–500; 5. 501–1,000; 6. 1,001–5,000; 7. 5,001–10,000; 8. 10,001–	4.527	2.307

Is your organization non-profit or for-profit?	25,000; 9. 25,001–50,000; 10. More than 50,000 1. Yes 0. No	0.366	0.482
In what business or industry is your establishment? (dummy variables)	1. Primary ⁶ 2. Secondary ⁷ 3. Tertiary ⁸ 4. Quaternary ⁹	0.041 0.194 0.624 0.141	0.199 0.396 0.485 0.348
<i>Workforce composition</i> What percentage of the permanent full-time and part-time employees in your organization are so-called white-collar employees?	0. 0%; 1. 1%-9%; 2. 10%-19%;; 10. 90%-99%; 11. 100%	5.566	3.576
About what percent of the employees at this establishment are aged 55 or over?	0. 0%; 1. 1%-9%; 2. 10%-19%;; 10. 90%-99%; 11. 100%	3.427	1.908
About what percent of the employees at this establishment work fewer than 35 hours per week?	0. 0%; 1. 1%-9%; 2. 10%-19%;; 10. 90%-99%; 11. 100%	4.577	2.630
About what percent of employees at this establishment are women?	0. 0%; 1. 1%-9%; 2. 10%-19%;; 9. 80%-89%; 10. 90%-99%; 11. 100%	4.977	2.458
About what percent of employees work in shift schedules?	0. 0%; 1. 1%-9%; 2. 10%-19%;; 9. 80%-89%; 10. 90%-99%; 11. 100%	2.041	2.838
<i>Non-standard work policies</i>			

⁶ Agriculture, forestry, and fishing; Mining and quarrying.

⁷ Industry; Production, distribution, trading of electricity and natural gas; Collection/distribution of water; waste(water) management, remediation; Construction industry.

⁸ Wholesale and retail trade; repair of cars; Transport and storage; Accommodation, meals and drinks; Financial Institutions; Renting and trading of real estate; Rental of movable property, other business service; Public administration, government services, social insurance; Health and Welfare; Culture, sport, and recreation; Other services; International organizations.

⁹ Information and communication; Consultancy, research, special business services; Education.

Think of a secure, full-time position in your establishment that is occupied by an employee in good standing. Either as a matter of formal or informal policy, would your establishment allow this employee flexible working times?	1. Yes 0. No	0.592	0.492
Think of a secure, full-time position in your establishment that is occupied by an employee in good standing. Either as a matter of formal or informal policy, would your establishment allow this employee job sharing, where two employees split one full-time job?	1. Yes 0. No	0.431	0.496

Table A3. Definition and measurement of explanatory variables on selected worker characteristics

Question	Response options	Mean	Std. dev.
<i>Demographic characteristics of selected worker</i>			
What is your best estimate of [Name]'s age?	Values from 55 to 70 years old	59.726	3.505
What is your best estimate of the highest level of schooling [Name] has completed?	1. Primary education 2. Lower secondary and pre-vocational education 3. Upper general secondary and pre-university education 4. Intermediate vocational education 5. Higher professional education 6. Academic university education	4.314	1.350
Do you know three [men/women] aged 55 or older who work full-time? (Gender of the randomly selected worker)	1. Male 0. Female	0.616	0.487
<i>Job performance of selected worker</i>			
[Name] easily learns new procedures and technologies	1. Completely disagree; 5. Completely agree	3.485	1.185

when they are introduced in the workplace			
[Name] is creative in solving problems on the job	1. Completely disagree; 5. Completely agree	3.892	1.042
[Name] is willing to put forth extra effort to get the job done	1. Completely disagree; 5. Completely agree	4.191	0.944
[Name] gets along well with co-workers	1. Completely disagree; 5. Completely agree	4.262	0.810
[Name] has the physical stamina or energy level needed to perform well in the job	1. Completely disagree; 5. Completely agree	3.904	1.008
<i>Occupation and workplace context</i>			
Which of the following occupational categories best describes [Name]'s position?	1. High-skill non-manual: Managers, Professionals	0.534	0.499
	2. Intermediate non-manual: Clerical support, Services and sales	0.270	0.445
	3. Skilled manual or technical: Skilled agriculture/forestry/fishery, Craft, Plant/machine operators; Low-skill or basic: Elementary occupations	0.196	0.397
Percentage of employees in the same position as the selected worker who work part-time	0. 0%; 1. 1%-9%; 2. 10%-19%;; 9. 80%-89%; 10. 90%-99%; 11. 100%	2.985	3.048
<i>Job characteristics of selected worker</i>			
Job of [Name] involves tasks that are discrete, in the sense that they have a clear beginning and end and are relatively self-contained	1. Completely disagree; 5. Completely agree	3.414	1.271
Job of [Name] requires that [she/he] performs most tasks alone	1. Completely disagree; 5. Completely agree	2.965	1.209
		4.325	0.875

Job of [Name] requires frequent or close communication with other members of a group	1. Completely disagree; 5. Completely agree	4.103	1.066
In general, [Name] job requires several years of experience to do well	1. Completely disagree; 5. Completely agree	3.559	1.428
About how many years has [Name] worked for your establishment?	1. Fewer than 5; 2. 5–10; 3. 11–15; 4. 16–25; 5. More 25	3.160	1.038
There are some jobs where, on average, older workers tend to be much more productive than younger workers. There are other jobs where this is not true. How would you rate the job of [Job title of selected worker] – on average, would you say older workers are:	1. much more productive than younger workers in this job? 2. moderately more productive than younger workers in this job? 3. just as productive as younger workers in this job? 4. moderately less productive than younger workers in this job? 5. much less productive than younger workers in this job?	2.752	0.778
Some jobs involve skills that need regular updating through employer-sponsored training. How often would you say that a typical individual in [Name]’s job receives employer-sponsored training – would you say often, sometimes, rarely, or never?	1. Never 2. Rarely 3. Sometimes 4. Often	2.100	1.115
If [Name] were to leave this job, it would be desirable to fill the job with a recent graduate	1. Completely disagree; 5. Completely agree	2.489	0.6125
Compared to 5 years ago, would you say it is now more difficult, less difficult, or about as difficult to find employees to fill positions like [Name]’s?	1: Less difficult 2: About as difficult 3: More difficult		

Table A4a. Ordered probit models explaining the likelihood that the selected worker would switch to a part-time position before the state pension age – Marginal effects

Characteristic of the selected worker	1 (Not at all likely)	2	3	4	5 (Very likely)
<i>Demographic characteristics of selected worker</i>					
Age	0.009* (0.004)	0.008 (0.006)	0.000 (0.001)	-0.002 (0.002)	-0.015* (0.009)
Education is higher	-0.004 (0.015)	-0.004 (0.013)	-0.000 (0.000)	0.001 (0.003)	0.007 (0.025)
Male	-0.127** (0.054)	-0.109 (0.073)	-0.003 (0.010)	0.027 (0.023)	0.212** (0.104)
<i>Job performance of selected worker</i>					
Learns new procedures and technologies	-0.047 (0.035)	-0.041 (0.029)	-0.001 (0.004)	0.010 (0.011)	0.079 (0.053)
Creative in solving problems	-0.019 (0.033)	-0.016 (0.026)	-0.000 (0.001)	0.004 (0.006)	0.031 (0.053)
Willing to make extra effort to get job done	0.084** (0.033)	0.072*** (0.021)	0.002 (0.007)	-0.018 (0.012)	-0.141*** (0.035)
Gets along well with coworkers	0.033 (0.022)	0.028 (0.018)	0.001 (0.003)	-0.007 (0.006)	-0.054 (0.034)
Has energy level needed to perform well in the job	-0.010 (0.031)	-0.009 (0.025)	-0.000 (0.001)	0.002 (0.006)	0.017 (0.050)
<i>Occupational category of selected worker</i>					
Intermediate non-manual	-0.271*** (0.069)	-0.250*** (0.084)	-0.020 (0.025)	0.088 (0.070)	0.453*** (0.096)
Skilled manual or technical; low-skill or basic	-0.300*** (0.072)	-0.327*** (0.092)	-0.065* (0.033)	0.002 (0.047)	0.690*** (0.069)
<i>Job characteristics of selected worker</i>					
<i>Job is compatible with part-time work</i>					
There are regular part-timers in selected worker's job title	-0.027*** (0.010)	-0.023** (0.012)	-0.001 (0.002)	0.006 (0.004)	0.045*** (0.016)
Job involves tasks that are discrete	-0.036 (0.024)	-0.031 (0.023)	-0.001 (0.003)	0.008 (0.007)	0.060 (0.039)
Most tasks are performed alone	-0.009 (0.021)	-0.007 (0.019)	-0.000 (0.001)	0.002 (0.005)	0.014 (0.036)
Job requires close communication with others	0.046 (0.043)	0.040 (0.025)	0.001 (0.004)	-0.010 (0.010)	-0.077 (0.059)
<i>Skill and training</i>	-0.054**	-0.046	-0.001	0.011	0.089*

Jobs where old are more productive than young	(0.022)	(0.036)	(0.004)	(0.010)	(0.049)
Selected worker's job tenure in establishment	0.053*** (0.017)	0.046** (0.022)	0.001 (0.004)	-0.011 (0.009)	-0.089*** (0.026)
Job requires several years of experience to do well	-0.011 (0.037)	-0.009 (0.031)	-0.000 (0.001)	0.002 (0.007)	0.018 (0.062)
Employer-sponsored training	0.083*** (0.030)	0.071** (0.030)	0.002 (0.007)	-0.017 (0.013)	-0.138*** (0.040)
<i>Difficulty in finding a replacement</i>					
Desirable to fill the job of worker with a recent graduate	0.065** (0.032)	0.056 (0.042)	0.001 (0.005)	-0.014 (0.014)	-0.109* (0.061)
Increased difficulty finding employees for selected worker's job type	-0.104** (0.047)	-0.089*** (0.034)	-0.002 (0.008)	0.022 (0.015)	0.174*** (0.058)
N	220				
Log pseudolikelihood	-184.982				
Pseudo R-squared	0.324				
p-value of Wald test for model significance	0.000				

Notes: 1. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$. 2. Standard errors based on the delta method. 3. Marginal effects may not sum exactly to zero due to rounding. 4. Base occupational category: High-skill non-manual.

Table A4b. Ordered probit models explaining the likelihood that the selected worker would switch to a part-time position after the state pension age – Marginal effects

Characteristic of the selected worker	1 (Not at all likely)	2	3	4	5 (Very likely)
<i>Demographic characteristics of selected worker</i>					
Age	-0.025** (0.011)	-0.002 (0.002)	-0.007 (0.005)	-0.010 (0.007)	0.044** (0.019)
Education is higher	-0.088*** (0.034)	-0.009 (0.008)	-0.025 (0.016)	-0.035 (0.022)	0.156*** (0.051)
Male	-0.181* (0.104)	-0.018 (0.018)	-0.050 (0.039)	-0.072* (0.043)	0.321** (0.155)
<i>Job performance of selected worker</i>					
Learns new procedures and technologies	-0.050 (0.039)	-0.005 (0.006)	-0.014 (0.012)	-0.020 (0.017)	0.088 (0.064)
Creative in solving problems	0.045	0.004	0.013	0.018	-0.080

Willing to make extra effort to get job done	(0.035) 0.045 (0.038)	(0.004) 0.004 (0.005)	(0.011) 0.013 (0.015)	(0.014) 0.018 (0.022)	(0.055) -0.080 (0.075)
Gets along well with coworkers	-0.013 (0.031)	-0.001 (0.003)	-0.004 (0.010)	-0.005 (0.014)	0.024 (0.058)
Has energy level needed to perform well in the job	-0.060* (0.032)	-0.006 (0.005)	-0.017 (0.015)	-0.024 (0.020)	0.107* (0.060)
<i>Occupational category of selected worker</i>					
Intermediate non-manual	-0.285*** (0.064)	-0.007 (0.008)	-0.020 (0.016)	-0.055 (0.036)	0.367*** (0.068)
Skilled manual or technical; low-skill or basic	-0.358*** (0.074)	-0.014 (0.013)	-0.036 (0.025)	-0.077** (0.036)	0.485*** (0.080)
<i>Job characteristics of selected worker</i>					
<i>Job is compatible with part-time work</i>					
There are regular part-timers in selected worker's job title	-0.025* (0.014)	-0.002 (0.002)	-0.007 (0.006)	-0.010 (0.008)	0.043* (0.026)
Job involves tasks that are discrete	-0.076*** (0.029)	-0.007 (0.007)	-0.021 (0.014)	-0.030* (0.018)	0.135*** (0.041)
Most tasks are performed alone	0.057** (0.027)	0.006 (0.006)	0.016 (0.011)	0.023* (0.013)	-0.102** (0.040)
Job requires close communication with others	0.060* (0.035)	0.006 (0.006)	0.017 (0.014)	0.024 (0.021)	-0.107* (0.064)
<i>Skill and training</i>					
Jobs where old are more productive than young	-0.090*** (0.034)	-0.009 (0.008)	-0.025 (0.019)	-0.036 (0.025)	0.160** (0.062)
Selected worker's job tenure in establishment	0.004 (0.027)	0.000 (0.003)	0.001 (0.008)	0.002 (0.011)	-0.007 (0.048)
Job requires several years of experience to do well	-0.095** (0.047)	-0.009 (0.009)	-0.026 (0.019)	-0.038* (0.022)	0.168** (0.070)
Employer-sponsored training	0.048 (0.032)	0.005 (0.005)	0.013 (0.011)	0.019 (0.014)	-0.086* (0.051)
<i>Difficulty in finding a replacement</i>					
Desirable to fill the job of worker with a recent graduate	0.008 (0.033)	0.001 (0.003)	0.002 (0.010)	0.003 (0.014)	-0.014 (0.060)
Increased difficulty finding employees for selected worker's job type	0.016 (0.051)	0.002 (0.005)	0.005 (0.015)	0.006 (0.022)	-0.029 (0.093)

N	180
Log pseudolikelihood	-142.623
Pseudo R-squared	0.337
p-value of Wald test for model significance	0.000

Notes: 1. *: $p < 0.10$; **: $p < 0.05$; ***: $p < 0.01$. 2. Standard errors based on the delta method. 3. Marginal effects may not sum exactly to zero due to rounding. 4. Base occupational category: High-skill non-manual.